

- 7 (a) In respiration, most ATP is synthesised during oxidative phosphorylation. Some ATP is made by substrate-linked reactions in glycolysis and Krebs cycle.

Describe how ATP is made by substrate-linked reactions.

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.....[2]

- (b) Lipids can be metabolised to provide ATP.

- The enzyme lipase hydrolyses lipids to glycerol and fatty acids.
- The hydrocarbon chain of the fatty acid breaks down into smaller, 2C compounds.
- Each 2C compound reacts with coenzyme A to form acetyl coenzyme A.

- (i) Name the covalent bond in lipids that is hydrolysed by lipase.

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- (ii) State the role of acetyl coenzyme A in respiration.

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- (iii) Explain why lipids have a higher energy value than carbohydrates.

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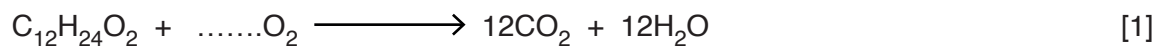
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- (c) The respiratory quotient, RQ, is used to show which substrate is being metabolised by cells. It can be determined using the equation below.

$$\text{RQ} = \frac{\text{molecules of carbon dioxide released}}{\text{molecules of oxygen taken in}}$$

Lauric acid is a saturated fatty acid found in coconuts and has a chain of 12 carbon atoms.

- (i) Complete the equation below which outlines the aerobic respiration of lauric acid.



- (ii) Calculate the RQ value for lauric acid.

Show your working. Give your answer to 2 decimal places.

answer = .....[2]

[Total: 9]